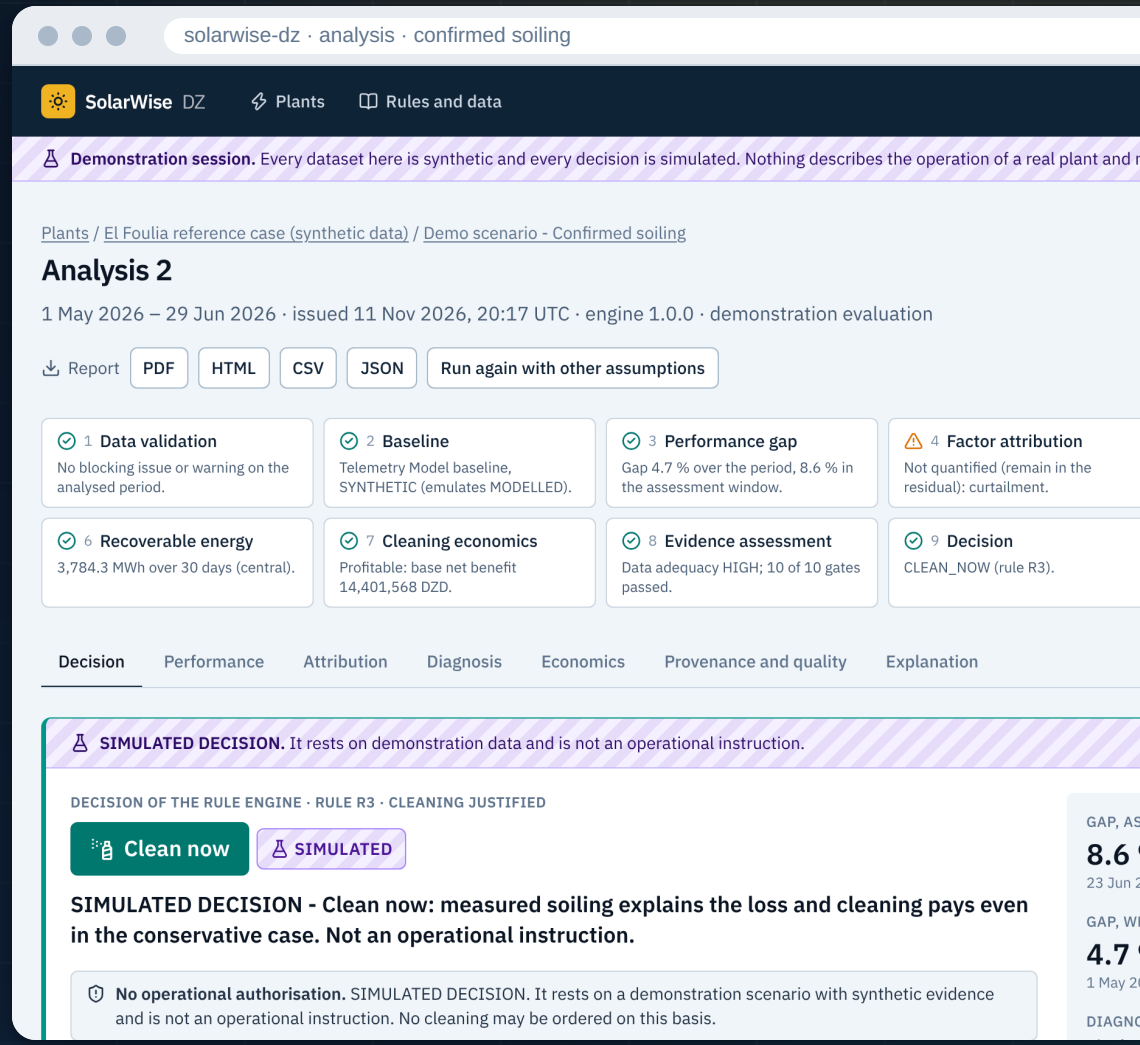
SolarWise DZ

Know why a solar plant under-performs before deciding to clean it.

Decision support for photovoltaic plants in the Algerian south. It explains the production gap, then tells whether a cleaning is justified.

Clean now | Investigate | Monitor



Reference case: El Foulia, 300 MWp, Guemar (El Oued), Algeria.
Every figure shown from the demonstration is synthetic.

THE PROBLEM

Production is down 9 %. Is it dust?

In the Sahara, dust really does cut output. But heat, a fault, shading, curtailment or bad data give the same picture on the meter.

−8.4 %
power after 8 weeks without cleaning, measured at Ouargla [1]

300 MWp
El Foulia (El Oued), Algeria's largest solar plant, inaugurated in October 2026 [2]

15,000 MW
of solar capacity planned by Algeria for 2035 [2]

ACTUAL AGAINST EXPECTED ENERGY · LAST 7 DAYS

−9 %



Six possible causes. One meter reading.

- Dust on the modules ?
- Heat ?
- Inverter fault ?
- Shading ?
- Grid curtailment ?
- Missing or wrong data ?

Clean for the wrong reason: water, labour and money are spent, and nothing is recovered.

Wait when it really is dust: energy is lost every day.

[1] Dida et al., Renewable and Sustainable Energy Reviews 124 (2020), field measurements at Ouargla. [2] pv magazine France, 28 October 2026.

OUR SOLUTION

SolarWise DZ answers three questions, in order.

It reads production data, soiling measurements, maintenance logs, inspections and your own costs, each with its declared origin.

1

How much is missing?

Actual energy against a documented baseline. The gap counts only when it exceeds the measurement uncertainty.

→ **The gap, with its uncertainty**

2

Why?

Each cause is quantified only from data that measure it: downtime, equipment, curtailment, soiling sensor. The rest stays “unexplained”.

→ **The causes, with their evidence**

3

Does cleaning pay?

The energy a cleaning would recover, at its value, against the full cost of cleaning. It must pay even in the conservative case.

→ **The net benefit, from low to high**

One answer, with its reasons

Clean now

Every gate is passed: the cleaning is justified and it pays.

Investigate

Something is unknown, open or contradictory. Find out first.

Monitor

Nothing to act on now. The analysis is repeated later.

QUESTION 2 · WHY?

Every kilowatt-hour of the gap is accounted for, or declared unexplained.

From expected to actual energy

SIMULATED

Scenario “confirmed soiling”, 23 to 29 June 2026 · gap −0.99 GWh (−8.6 %)



8.5 % **Measured** by the soiling sensor. It is not deduced from the gap.

8.4 % **Left for soiling to explain** once the equipment losses are removed. The two agree.

0.1 % **Unexplained.** Nothing is hidden inside “soiling”.

Sunshine and temperature are already inside the expected energy. Performance metrics follow IEC 61724-1.

Synthetic demonstration scenario on the El Foulia reference case. It contains no measurement from any real plant.

QUESTION 3 · THE DECISION

A cleaning order has to pass ten gates.

THE DATA	1 Reliable measured production data	2 Baseline of documented origin
THE LOSS	3 Significant under-performance now, beyond the uncertainty	
THE SOILING	4 Credible measured soiling evidence	5 Soiling consistent with the production gap
NOTHING ELSE	6 No unresolved technical issue	7 No critical contradiction in the evidence
THE MONEY	8 Recoverable energy estimated	9 Complete economic inputs · 10 Profitable in the conservative case

10 of 10

Clean now

Measured loss, explained by measured soiling, nothing else open, and a cleaning that pays.

Anything less

Investigate **Monitor**

A gate that cannot be evaluated counts as not passed.

Explicit rules: same data, same answer.

Every result is stored as issued.

An independent interlock re-checks every “Clean now”.

WHAT IT CHANGES

Six cases with the same 9 % drop. One cleaning order.

−8.6 % — Clean now

Dust, measured

The soiling sensor reads 8.5 %. It matches the gap, and the cleaning pays.

−8.9 % — Monitor

Heat

Module temperature explains 91 % of the gap. Cleaning would change nothing.

−10.6 % — Investigate

Inverter fault

Part of the plant is not producing. Repair first: cleaning would not recover it.

−9.2 % — Investigate

Morning shading

The sensor shows a clean array and 8.7 % stays unexplained. Check on site.

−10.1 % — Monitor

Grid curtailment

The grid limited the output. Cleaning would not bring that energy back.

−8.6 % — Investigate

Conflicting evidence

An inspection reports heavy soiling, the sensor reads 1 %. Resolve it first.

A rule of thumb, “gap above 5 %: clean”, would have sent the cleaning crews six times.

SIMULATED

Gap over the last 7 days of each scenario

Synthetic demonstration scenarios on the El Foulia reference case. They contain no measurement from any real plant.

DOES CLEANING PAY?

Same cleaning cost. Opposite answers.

Dry weeks ahead

Soiling measured at 8.5 %.
Benefit counted over 30 days.

Clean now

Cost of the cleaning	4.5 M DZD
Energy recovered	18.9 M DZD

+14.4 M DZD

Net benefit. Still +5.0 M DZD in the conservative case.

10 gates of 10 passed. The cleaning is justified.

Rain expected within days

Soiling measured at 5.4 %.
Benefit counted over 3 days.

Monitor

Cost of the cleaning	4.5 M DZD
Energy recovered	1.2 M DZD

−3.3 M DZD

Net loss. Still −1.1 M DZD in the optimistic case.

9 gates of 10 passed. Gate 10 fails: the cleaning does not pay.

The test is conservative: low recoverable energy against high cost. The net benefit must stay positive.

No price is built in. The operator enters water, labour, equipment, downtime and the value of a kWh.

Simulated scenarios with illustrative assumptions (5 DZD per kWh). These are not market prices.

WHY YOU CAN TRUST THE ANSWER

Four rules the software never breaks.

A gap is not a diagnosis

Lower production is an observation. It never proves soiling by itself.

Every value has an origin

Measured, modelled, synthetic or unknown: declared with the data and shown with every figure. Only measured data can justify a cleaning.

Rules decide. AI only explains.

A deterministic rule engine decides. An optional language model may only rephrase the explanation, and its text is checked against the engine's figures.

Missing stays missing

A value that was not observed is never replaced by zero and never interpolated.

IN THE APPLICATION

Every decision shows the origin of the evidence it rests on.

Evidence the decision rests on

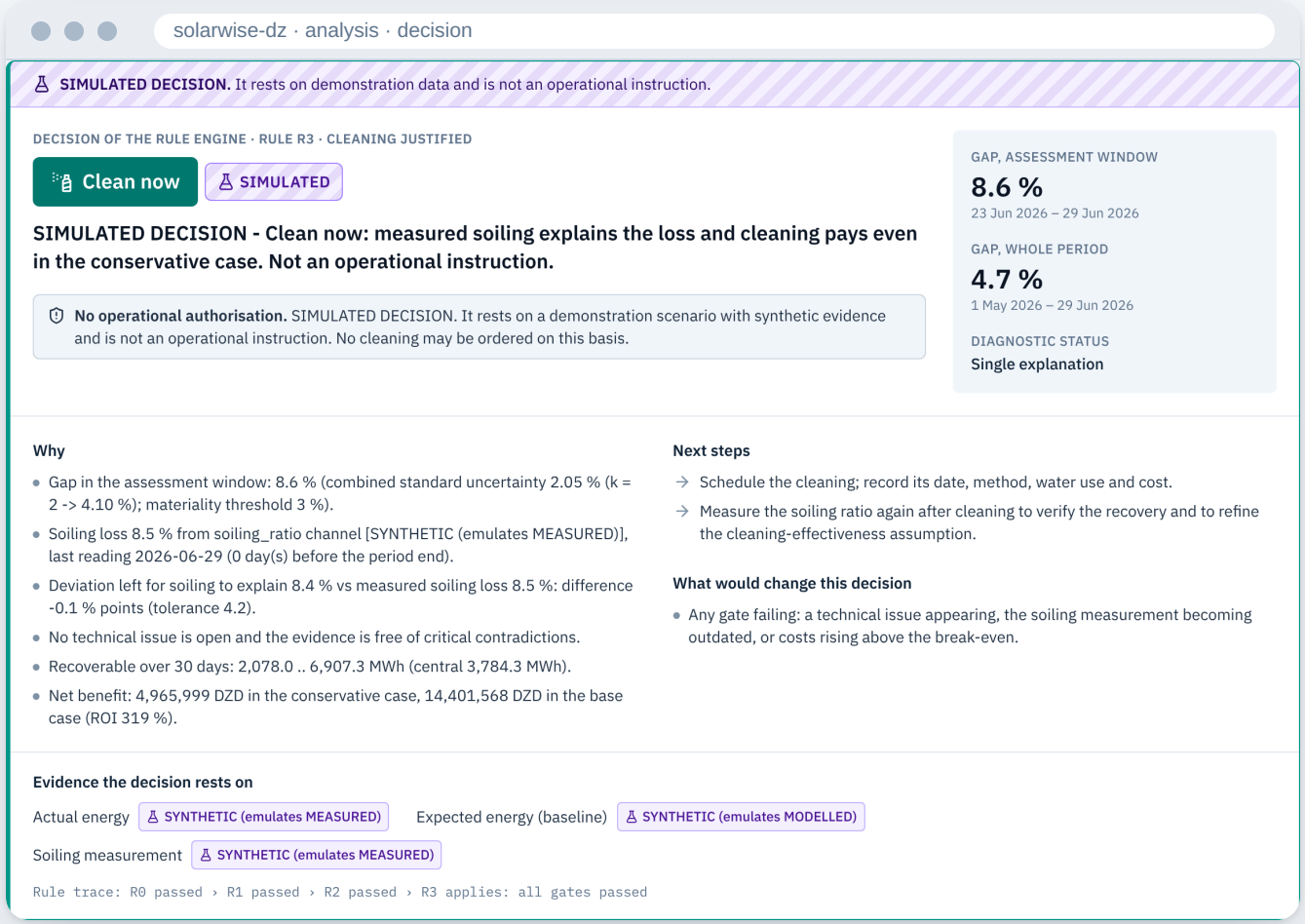
Actual energy — SYNTHETIC (emulates MEASURED)

Expected energy (baseline) — SYNTHETIC (emulates MODELLED)

Soiling measurement — SYNTHETIC (emulates MEASURED)

WHERE WE ARE

A working application, not a mock-up.



10 — **demonstration scenarios**, each one giving the decision it should

871 — **automated tests**: 788 on the engine and the API, 83 on the interface

43 — **API operations** behind a web interface, with accounts and roles

4 — **report formats**: PDF, HTML, CSV and JSON. Every result is stored as issued

Validated on synthetic scenarios. It has not yet been run on the data of a real plant.

NEXT STEPS

From demonstration to a pilot plant.

1

Pilot

Run on the real production exports of one plant.

2

Calibrate

Tune thresholds and uncertainties with the operator's engineers.

3

Connect

Soiling stations and weather feeds. The PVGIS and NASA POWER connectors are already built.

4

Scale

More plants, an interface in Arabic and French, planning of cleaning campaigns.

WHAT WE ARE LOOKING FOR

One plant's historical data, and an operator ready to test it with us.

WHAT THE OPERATOR GETS

An evidence file for every cleaning decision: what was measured, what was assumed, and whether the cleaning pays.

SolarWise DZ

Clean when the evidence says so.

How much is missing. Why. Whether cleaning pays.

Clean now | Investigate | Monitor

Thank you.

Demonstration on synthetic data · El Foulia reference case, 300 MWp, El Oued

BACKUP · THE DECISION RULES

The rules are applied in this order. The first that applies decides.

RULE	APPLIES WHEN	OUTCOME	WHY
R0	Production data are not reliable measured data, or the baseline is not documented.	Investigate	Without trustworthy production and a known reference there is no gap to reason about.
R1	A technical issue is open in the assessment window.	Investigate	Its losses cannot be separated from soiling, and cleaning would not recover them.
R2	The evidence contains a critical contradiction.	Investigate	A decision resting on one of two contradicting sources would be arbitrary.
R3	Every one of the ten gates is passed.	Clean now	Measured loss, explained by measured soiling, nothing else open, and a cleaning that pays.
R4	The deviation is significant and largely unexplained, and soiling is not credibly measured.	Investigate	A production gap alone does not prove soiling.
R5	Soiling is measured and matches the deviation, but the economic inputs are not decision-grade.	Investigate	It cannot yet be shown that cleaning pays.
R6	None of the above: no significant deviation, a cause that cleaning does not change, or a cleaning that would not pay.	Monitor	Nothing to act on now. The analysis is repeated after the review interval.
Interlock	A “Clean now” is checked a second time, independently, against the gates and the origin of its evidence, and the check fails.	Investigate	A programming error in the rules must not be able to issue a cleaning order.

BACKUP · ARCHITECTURE

From a file to a decision, in five steps.

Data in

CSV or JSON files. Each column is declared: unit, origin, source.

Validation

Units, gaps, duplicates, completeness. Missing stays missing.

Engines

Gap, attribution, diagnosis, recoverable energy, economics.

Rule engine

Ten gates, rules R0 to R6, and the safety interlock.

Results

Decision, explanation, reports in PDF, HTML, CSV and JSON.

BUILT WITH	Python, FastAPI, SQLAlchemy, PostgreSQL or SQLite · React, TypeScript, ECharts
OPTIONAL CONNECTIONS	PVGIS typical year, NASA POWER weather, a language model that rephrases the explanation. All can stay switched off.
ACCESS	Sign-up, which can be switched off. Roles for each plant: owner, editor, viewer. Sign-ins and changes are recorded.
VERIFIED BY	788 tests on the engine and the API, 83 on the interface, static typing and linting on both sides.

BACKUP · DEMONSTRATION

The ten demonstration scenarios and what the engine answers.

#	SCENARIO	GAP, LAST 7 DAYS	DECISION	RULE AND REASON
1	Normal operation	0.5 %	Monitor	R6 · production is in line with the baseline
2	Confirmed soiling	8.6 %	Clean now	R3 · every gate is passed
3	Temperature effect	8.9 %	Monitor	R6 · explained by a cause that cleaning does not address
4	Inverter fault	10.6 %	Investigate	R1 · a technical issue is open
5	Morning shading	9.2 %	Investigate	R4 · a significant loss is not explained
6	Grid curtailment	10.1 %	Monitor	R6 · explained by a cause that cleaning does not address
7	Missing data	6.5 %	Investigate	R0 · the data are not sufficient to support a decision
8	Conflicting evidence	8.6 %	Investigate	R2 · the evidence contradicts itself
9	Synthetic data, no role declared	8.6 %	Investigate	R0 · the production data are not measured data
10	Unprofitable cleaning	5.8 %	Monitor	R6 · soiling is measured, but cleaning now would cost more than it recovers

Synthetic scenarios, 1 May to 29 June 2026, generated from a fixed seed. Every decision is marked simulated in the application.

BACKUP · LIMITS

What the solution does not do yet, stated plainly.

Tested on synthetic scenarios only

No measurement from El Foulia or from any other plant has been used. The name, the location and the nominal capacity are public information; everything else is an illustrative assumption.

Thresholds are documented defaults

The 7-day window, the 2 % floor for a credible soiling reading and the other thresholds must be calibrated with each operator.

No direct soiling measurement, no cleaning order

Without a soiling station, a clean reference module or an equivalent measurement, the answer is “Investigate”. This is by design.

The projections are not a validated forecast

The application shows simple reference projections for orientation. They never enter a decision.

Weather services are context, not evidence

PVGIS and NASA POWER data help to build a baseline. They do not measure the dust on the modules.